# AI POWERED JAVA FULLSTACK WITH KAFKA & AWS COURSE

## BACK-END DEVELOPMENT

### CORE JAVA

- **Introduction to Java**
  - History and Features of Java
  - JDK, JRE, JVM
  - Setting Up Java Environment and Eclipse
- **Basics of Java**
  - Structure of a Java Project
  - Creating First Java Project
  - Command Line Arguments
  - Primitive Data Types
  - Types of Variables
  - Naming Conventions
  - Operators
  - Control Statements
  - Loops
  - Continue and Break Statements
  - Type Casting
  - String
  - Arrays
  - Methods in Java
- **Object-Oriented Programming (OOP) Concepts**
  - Class and Object
  - Constructor
  - Constructor Chaining
  - Static Keyword
  - Inheritance
  - This vs Super
  - Polymorphism
  - Abstraction
  - Encapsulation
  - Access Modifiers
  - Interfaces
- **Miscellaneous**
  - Garbage Collection
  - File Handling
  - Final Keyword
  - Final vs Finalize
- **Exception Handling**
  - Types of Exceptions
  - Try, Catch, Finally
  - Throw vs Throws
  - Custom Exceptions
- **Features of java.lang Package**
  - Object Class
  - System Class
  - Math Class
  - Wrapper Classes
  - String Buffer vs String Builder
- **Multithreading**
  - What is a Thread in Java
  - Thread Lifecycle
  - Multithreading Advantages
  - Creating Threads using Runnable Interface & Thread Class
  - Synchronized Keyword
- **Java Collections**
  - Collections Overview
  - List, Set, Map Interfaces
  - List Implementations
  - Set Implementations
  - Map Implementations
  - Comparator and Comparable Interfaces
- **Java 8 Features**
  - Lambda Expression
  - Functional Interfaces
  - Stream API
  - Stream Filter
  - Changes in Interfaces - Default and Static Methods
  - forEach Method
  - Optional Classes
  - Date and Time API

# DATABASE

- **Structured Query Language (SQL)**
  - MySQL Environment Setup
  - Introduction to RDBMS
  - DDL (Data Definition Language)
  - DML (Data Manipulation Language)
  - DQL (Data Query Language)
  - DCL (Data Control Language)
  - TCL (Transaction Control Language)
  - Commit and Rollback
  - Constraints (Unique Key, Not Null, Primary, Foreign Key)
  - Aggregate Functions (MIN, MAX, AVG, SUM, COUNT)
  - Joins

# HIBERNATE

- **Introduction to Hibernate**
  - What is ORM?
  - JDBC vs Hibernate
- **Hibernate Core Concepts**
  - Configuring Hibernate Configuration XML File
  - Understanding Entity Classes
  - Id Annotation and Generation Methods
  - Cascade Types
- **Manipulating and Querying Persistent Data**
  - Session Factory
  - Session
  - CRUD Operations
  - Transaction Management
- **Hibernate Relationships & Mapping**
  - One-to-One Mapping
  - One-to-Many Mapping
  - Many-to-One Mapping
  - Many-to-Many Mapping
  - Eager vs Lazy Loading
- **Advanced Hibernate**
  - Hibernate Query Language (HQL)
  - Native Queries
  - Named Queries

# SPRING FRAMEWORK

- **Introduction to Spring**
  - What is a Framework?
  - Advantages of Spring
  - Spring Modules Overview
- **Spring Core**
  - What are Beans?
  - Spring Containers
  - Bean Lifecycle
  - Understanding IOC & DI
  - Bean Scope
  - Singleton Design Pattern
  - Spring Bean Configuration using XML
  - Java-based Spring Bean Configuration
  - Annotation-based Spring Bean Configuration
  - Types of Dependency Injection
- **Spring JDBC**
  - Configuring Data Source
  - Configuring JDBC Template
  - CRUD Operations using JDBC Template
  - Row Mapper
- **Spring ORM**
  - Advantages of Spring ORM over Hibernate
  - Configuring LocalSessionFactoryBean

- Configuring HibernateTemplate
- CRUD Operations using HibernateTemplate

- **Spring MVC**
  - Understanding MVC Architecture
  - Dispatcher Servlet
  - Types of Controllers
  - Model and View
  - Internal View Resolver
  - Working with JSPs and HTML Pages
- **Spring AOP**
  - Overview of Spring AOP
  - Join Points, Point Cuts, Advice, Aspects
  - Logging using AOP

## SPRING BOOT

- **Introduction to Spring Boot**
  - Features and Advantages of Spring Boot
  - Spring vs Spring Boot
  - Auto Configuration in Spring Boot
  - Creating Spring Starter Projects
  - Understanding application.properties and application.yml Files
  - Spring Boot Starters
  - Annotations in Spring Boot
- **Spring Boot Rest**
  - Building Rest APIs
  - RestController vs Controller
  - Handling HTTP Methods
  - Handling Status Codes using ResponseEntity
  - Working with Thymeleaf
- **Spring Data JPA**
  - Spring Data JPA Overview
  - CRUD Operations with JPA Repository
  - Finder Methods in JPA
  - Java Persistence Query Language (JPQL)
  - Custom Queries
  - @Transactional Annotation
- **Exception Handling**
  - RestControllerAdvice and ControllerAdvice
  - Exception Handlers
- **Spring Boot Actuator**
  - What is an Actuator?
  - Metrics using Actuator
- **Scheduling**
  - What is scheduling
  - Advantages of scheduling
  - What are Cron jobs?
  - Enabling scheduling in spring boot with cron expressions.

## REDIS CACHING

- **Introduction to Caching**
  - What is caching?
  - Why caching is needed (DB bottleneck problem)
  - Types of caching:
  - In-memory vs Distributed cache
- **Introduction to Redis**
  - What is Redis? (In-memory data store)
  - Key features:

- High performance
- Key-value store

- **Redis Setup**
  - Installing Redis (local setup)
  - Running Redis server
  - Redis CLI basics.
- **Spring Boot Integration with Redis**
  - Adding Redis dependency
  - Configuration (application.yml)
  - Using RedisTemplate
- **Caching with Spring Boot**
  - @Cacheable
  - @CachePut
  - @CacheEvict
- **TTL (Time To Live)**
  - Expiration of cache
  - Setting TTL in Redis
  - Why TTL is important

## MICROSERVICES

- **Introduction to Microservices**
  - Monolithic vs Microservices Architecture
  - Advantages & Challenges of Microservices
- **Service Discovery**
  - What is Service Discovery?
  - Creating Eureka Server
  - Registering Services in Eureka
- **Inter-Service Communication**
  - Synchronous vs Asynchronous Communication
  - Configuring Rest Template
  - Communicating using Rest Template
  - Configuring Feign Client
  - Communicating using Feign Client
- **Config Server**
  - What is a Config Server?
  - Creating an In-Memory Config Server
  - Config Server with GitHub
- **API Gateway**
  - What is a Gateway?
  - Setting Up an API Gateway
  - Routing Requests to Services
- **Circuit Breaker**
  - Why Do We Need a Circuit Breaker?
  - Setting Up Resilience4j
  - Writing Fallback Methods
  - Setting Thresholds and Managing Circuit

## KAFKA

- **Messaging :**
  - What is Asynchronous messaging?
  - Synchronous messaging vs Asynchronous messaging?
  - When Should I use Async?
- **Intro to Kafka Components:**
  - What is a kafka broker?
  - Producer vs Consumer in kafka
  - What are partitions in kafka

- What are topics in Kafka
- What are offsets in kafka

- **Kafka Integration With SpringBoot:**
  - What is a kafka template
  - Setting up a Spring Boot project with Kafka.
  - Publishing a message into a kafka topic using Spring Boot.Introduction to caffeine cache.
  - Consuming a message using Kafka Listeners.
- **Serialization & Deserialization:**
  - StringSerializer vs JSON
  - Using JSON Serialisation and Deserialisation in Kafka.

## CLOUD & DEPLOYMENT

- **Cloud Computing**
  - What is cloud computing?
  - On prem vs Cloud
- **Introduction to AWS**
  - What is AWS?
  - Setting up your first AWS account.
  - AWS Free Tier overview
  - Regions & Availability Zones
  - AWS Console walkthrough
- **Simple Storage Service**
  - Intro to s3 buckets.
  - Create an S3 bucket.
  - Configure permissions (public access)
  - Integrate S3 with spring boot application.
  - How to Put a file into S3 buckets.
  - How to get a file from s3 buckets.
  - What is a Pre-signed URL?
  - Uploading and downloading using a pre-signed url.
- **Deployment**
  - What is Deployment?
  - Local Vs Cloud deployment
  - What happens when we hit a URL:
  - Browser → DNS → Server → Response
- **Deployment using EC2**
  - What is EC2 (virtual machine)
  - Launch a new instance.
  - Configure Amazon Machine Image.
  - Connecting to EC2:
  - SSH / terminal
- **Networking & Security Basics**
  - What is Security Group (Firewall)
  - Important ports:
  - 22 (SSH)
  - 8080 (Spring Boot)
  - Public IP vs Private IP.
- **Deploy a Full stack Application in EC2.**
  - Install and setup Java and Maven in EC2.
  - Generate Spring Boot Jar.
  - Upload Spring Boot JAR to EC2 instance
  - Run application.

- Run application.
- Accessing backend APIs through public IP.
- Generating a public url using public DNS.
- Deploying a frontend application in Netlify.
- API integration from Front end to Backend using public DNS.

# SELF - PACED MODULES

## DATABASE

- **NoSQL**
  - MongoDB Environment Setup
  - Basic Overview of MongoDB
  - SQL vs NoSQL
  - Creating and Dropping Databases
  - Creating and Dropping Collections
  - Inserting, Updating, and Deleting Documents

## JDBC

- **JDBC (Java Database Connectivity)**
  - Understanding Java Database Connectivity
  - Driver Manager
  - Connection
  - Statement
  - Prepared Statement
  - Result Set Metadata
  - CRUD Operations
  - ResultSet

## WEB APPLICATIONS

- Building a Dynamic Web Application
- Setting Up Tomcat Server
- Initializing a Servlet
- Servlet Life Cycle
- doPost and doGet Methods
- Overview of JSPs
- Directives and Expressions
- Using JSPs with Servlets

## UNIT TESTING

- **JUnit**
  - Importance of Unit Testing in Software Development
  - Setting Up JUnit in Java
  - Annotations in JUnit
  - Assertions in JUnit
  - What is Mockito?
  - Mocking Database using Mockito
  - Performing Tests on Service Layers

## GIT & GITHUB

- Setting Up Git Environment
- What is a Version Control System?
- Why Use Git?
- Different Stages in Git
- Cloning a Repo using Git Clone
- Commands in Git
- Viewing Commit History
- What is Branching?
- Creating, Deleting & Switching Branches
- Creating a Pull Request
- Reviewing and Merging Pull Requests
- Resolving Merge Conflicts
- Sending custom objects

# FRONTEND

## Module 1: HTML & Basic Web Structure

- Introduction: Frontend vs Backend
- HTML Document Structure & Boilerplate
- Text Elements: Headings, Paragraphs, Lists
- Media Elements: Images, Links
- Semantic HTML: header, footer, section, article
- Forms & Input Types
- Accessibility Basics (Alt text, Labels, Landmarks)

## Module 2: CSS & Responsive Web Design

- CSS Syntax & Selectors
- Units, Colors, Typography
- Box Model (Margin, Padding, Border)
- Flexbox Layout
- CSS Grid Layout
- Responsive Design & Media Queries
- Transitions & Animations
- UI Styling: Buttons, Cards, Navigation Bars

## Module 3: Bootstrap

- What is Bootstrap?
- Using bootstrap
- Container layout
- Buttons
- Badges
- Alert,Button Group
- Navbar
- Card components
- Form layout

## Module 4: JavaScript Essentials

- Variables & Data Types (let, const, var)
- Operators & Expressions
- Conditionals & Loops
- Functions: Declaration, Expression, Arrow
- Arrays & Objects Manipulation
- DOM Selection & Manipulation
- Events & Event Listeners
- Form Validation
- ES6+ Features (Spread, Rest, Template Literals)

## Module 5: Advanced JavaScript

- Closures & Lexical Scope
- Hoisting
- Higher-Order Functions (map, filter, reduce)
- Debouncing & Throttling
- Asynchronous JS (Promises, async behavior)
- Fetch API & JSON Handling
- Browser Storage: localStorage

## Module 6: React.js Fundamentals

- What is React & Why React?
- JSX & Component Structure
- Functional Components & Props
- useState Hook
- Event Handling & Controlled Inputs
- useEffect Hook
- Rendering Lists & Keys
- Conditional Rendering
- Component Reusability Principles

## Module 7: Advanced React

- State Lifting & Prop Drilling
- Custom Hooks
- useRef & DOM Handling
- useMemo & Performance Optimization
- React.memo for Re-Rendering Control
- Folder Structure & Best Code Practices

## Module 8: Redux Toolkit

- Need for State Management & Redux Basics
- Store Setup with configureStore
- Slices with createSlice
- Actions & Reducers
- React Integration: Provider, useDispatch, useSelector
- Async Handling using createAsyncThunk

## Module 9: Deployment & Best Practices

- React App Build & Deploy
- Netlify / Vercel Deployment Basics
- Code Splitting Concepts
- Basic SEO & Accessibility in Frontend

# PROJECTS

## MAJOR PROJECTS

1 . MedSync - AI-Powered Hospital Management System
2 . StayEase - Accommodation Management System
3 . Smart Bank Management System
4 . PFMS-Personal Finance Management System
5 . Spotify Clone – Music Streaming UI
6 . ShopVibe E-commerce
7 . Blog Application

## MINI PROJECTS

1 . Music Player Application
2 . Online Food Ordering System
3 . Gated Community - Complaint Tracker Application
4 . AI-Based Resume Screening System using Spring AI
5 . Portfolio Grid Gallery
6 . Personal Portfolio Website
7 . Product Showcase Carousel

# WHY LEARN THIS COURSE

- This course is for those who want to start their coding journey, whether a beginner or an expert.
- No prior programming knowledge is required.
- This course covers basics to advanced concepts straightforwardly and systematically.
- We take special care of Non-IT students
- Easy to learn
- Beginner Friendly
- The diversity of our learners adds richness to course discussions and interactions.



# DELIVERABLES

10.DOWNLOADABLE RESOURCES

1.FROM SCRATCH TO MASTER LEVEL TRAINING

9.ON-DEMAND VIDEO COURSE CONTENT

2.RESUME BUILDING

8.PLACEMENT UPDATES

3.DAILY ASSIGNMENTS

7.Q&A SESSIONS

4.LINKEDIN PROFILE BUILDING

6.COURSE COMPLETION CERTIFICATE

5.INTERVIEW GUIDANCE

## TOP CITIES FOR AI POWERED JAVA FULLSTACK WITH KAFKA & AWS

- BANGALORE (BENGALURU)
- GURGAON (GURUGRAM)
- HYDERABAD
- PUNE
- MUMBAI
- CHENNAI
- NOIDA

## KEY SECTORS DRIVING DEMAND FOR AI POWERED JAVA FULLSTACK WITH KAFKA & AWS

- TECHNOLOGY AND SOFTWARE DEVELOPMENT
- E-COMMERCE HEALTHCARE
- FINANCE AND FINTECH
- HEALTHCARE
- EDUCATION AND EDTECH
- DATA SCIENCE AND ANALYTICS
- ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
- MEDIA AND ENTERTAINMENT
- STARTUPS AND TECH INCUBATORS

# TOP COMPANIES HIRING



























# ALL THE BEST

